

Day 5: Bren Math Workshop

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Derivatives practice

Summary of differentiation rules

Differentiation Rule	$f'(x)$ notation	$\frac{d}{dx}$ notation
Constant Rule	$f(x) = c \Rightarrow f'(x) = 0$	$\frac{d}{dx}[c] = 0$
Power Rule	$f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$	$\frac{d}{dx}[x^n] = nx^{n-1}$
Sum Rule	$(g(x) + h(x))' = g'(x) + h'(x)$	$\frac{d}{dx}[g(x) + h(x)] = \frac{d}{dx}g(x) + \frac{d}{dx}h(x)$
Difference Rule	$(g(x) - h(x))' = g'(x) - h'(x)$	$\frac{d}{dx}[g(x) - h(x)] = \frac{d}{dx}g(x) - \frac{d}{dx}h(x)$
Constant Multiplier Rule	$(c \cdot g(x))' = c g'(x)$	$\frac{d}{dx}[c g(x)] = c \frac{d}{dx}g(x)$

And two special derivatives: $\frac{d}{dx}(e^x) = e^x$ and $\frac{d}{dx}\ln(x) = \frac{1}{x}$.

There's more differentiation rules (product, quotient, chain rule).

They're fun, but we won't be reviewing them.

Derivatives practice

Solve individually and then check with a partner.

1. Find the derivatives of these functions

A) $y = 3x^2$

B) $h(x) = 2x(x^2 + 3x)$

C) $g(y) = 2\sqrt{y} - e^y + 20 \ln(y)$

2. A city's community solar program has installed cumulative solar capacity, $S(t)$ (in megawatts = MW), modeled as:

$$S(t) = 0.5t^2 + 4t, \text{ where } t \text{ is years since 2015.}$$

Derivatives solution

1. Find the derivatives of these functions

$$\text{A) } y = 3x^2 \qquad \text{B) } h(x) = 2x(x^2 + 3x)$$

$$\text{a. } y' = 3 \cdot 2x^{(2-1)} = 6x$$

$$\text{b. First expand: } h(x) = 2x \cdot x^2 + 2x \cdot 3x = 2x^3 + 6x^2$$

$$\text{Then differentiate term by term: } h'(x) = 2 \cdot 3x^{(3-1)} + 6 \cdot 2x^{(2-1)} = 6x^2 + 12x$$

Derivatives solution

1,C) Find the derivative of this function

$$g(y) = 2\sqrt{y} - e^y + 20 \ln(y)$$

Rewrite $\sqrt{y}$ as $y^{1/2}$:

$$\begin{aligned} g(y) &= 2y^{1/2} - e^y + 20 \ln(y) \\ g'(y) &= 2 \cdot \frac{1}{2} y^{(1/2-1)} - e^y + 20 \cdot \frac{1}{y} \\ &= \frac{1}{\sqrt{y}} - e^y + \frac{20}{y} \end{aligned}$$

Derivatives solution

3. Since 2015, a city's community solar program has installed cumulative solar capacity, $S(t)$ (in MW), modeled as:

$$S(t) = 0.5t^2 + 4t$$

where t is years since 2015.

a. Evaluate $S(6)$. What does this value mean in context?

We have that $S(6) = 0.5(6)^2 + 4 * 6 = 42$. This means that in 2021 (6 years after the program began), the city's installed solar capacity was 42 MW.

b. Find $S'(t)$.

$$S'(t) = t + 4$$

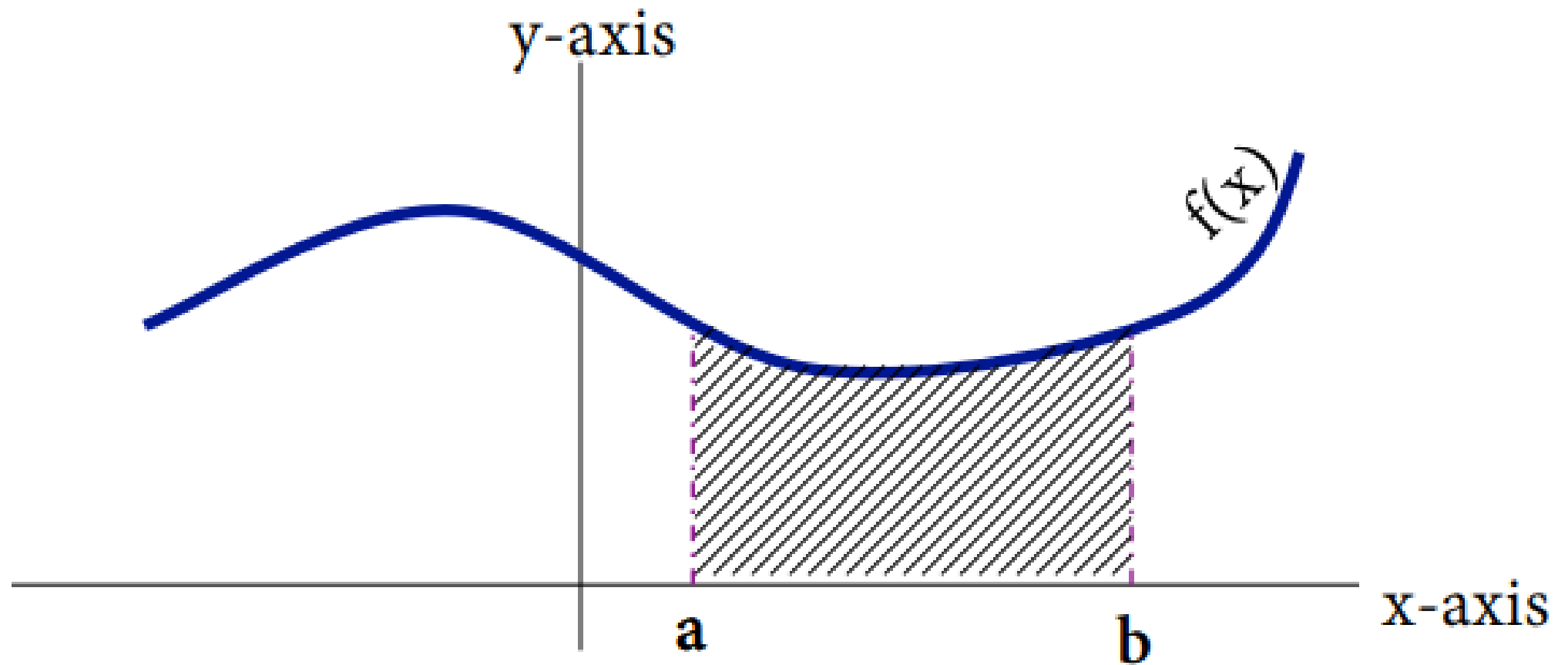
c. Evaluate $S'(6)$. What does this value mean in context?

$$S'(6) = 6 + 4 = 10$$

In 2021 (6 years after the program began), the city's installed solar capacity was increasing at a rate of 10 MW per year.

Integration Conceptually

How do we find the area under this curve?



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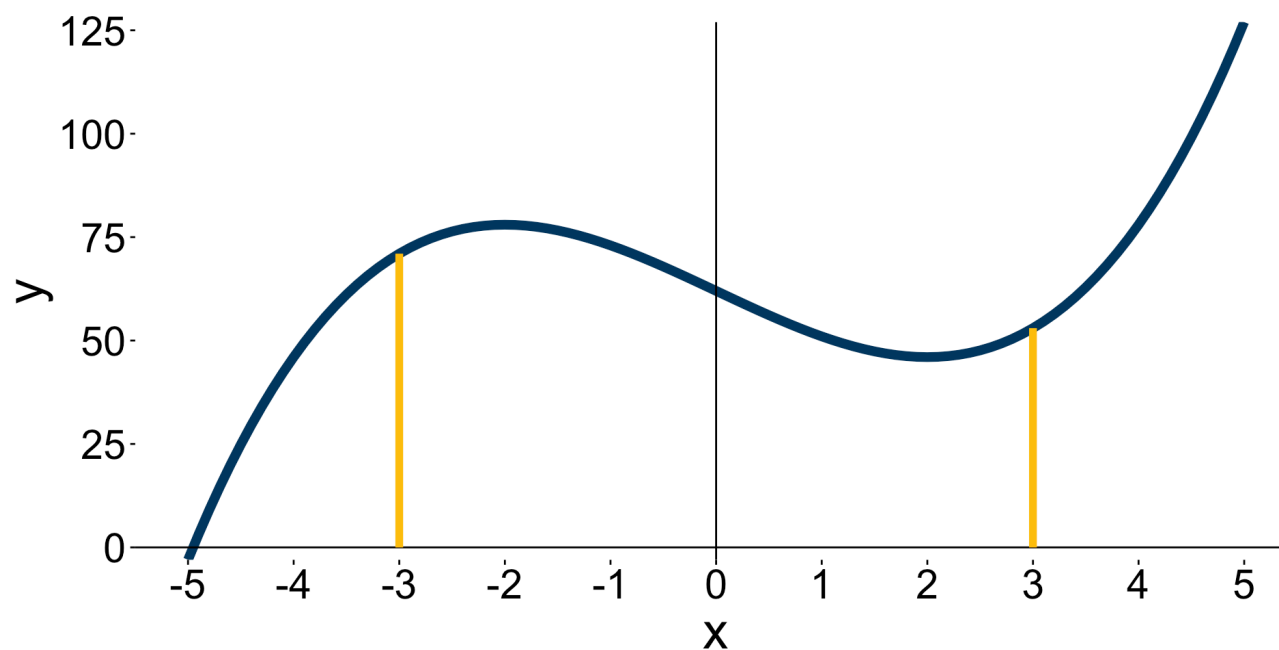
Areas of rectangles are easy to find. We can approximate the area by adding the area of rectangles under the curve.

Example

Approximate the area under the curve from $[-3, 3]$ with $\Delta x = 2$ and $k = 3$.



x	y
-3	71
-1	73
1	51
3	53



Example

Approximate the area under the curve from $[-3, 3]$ with $\Delta x = 2$ and $k = 3$.

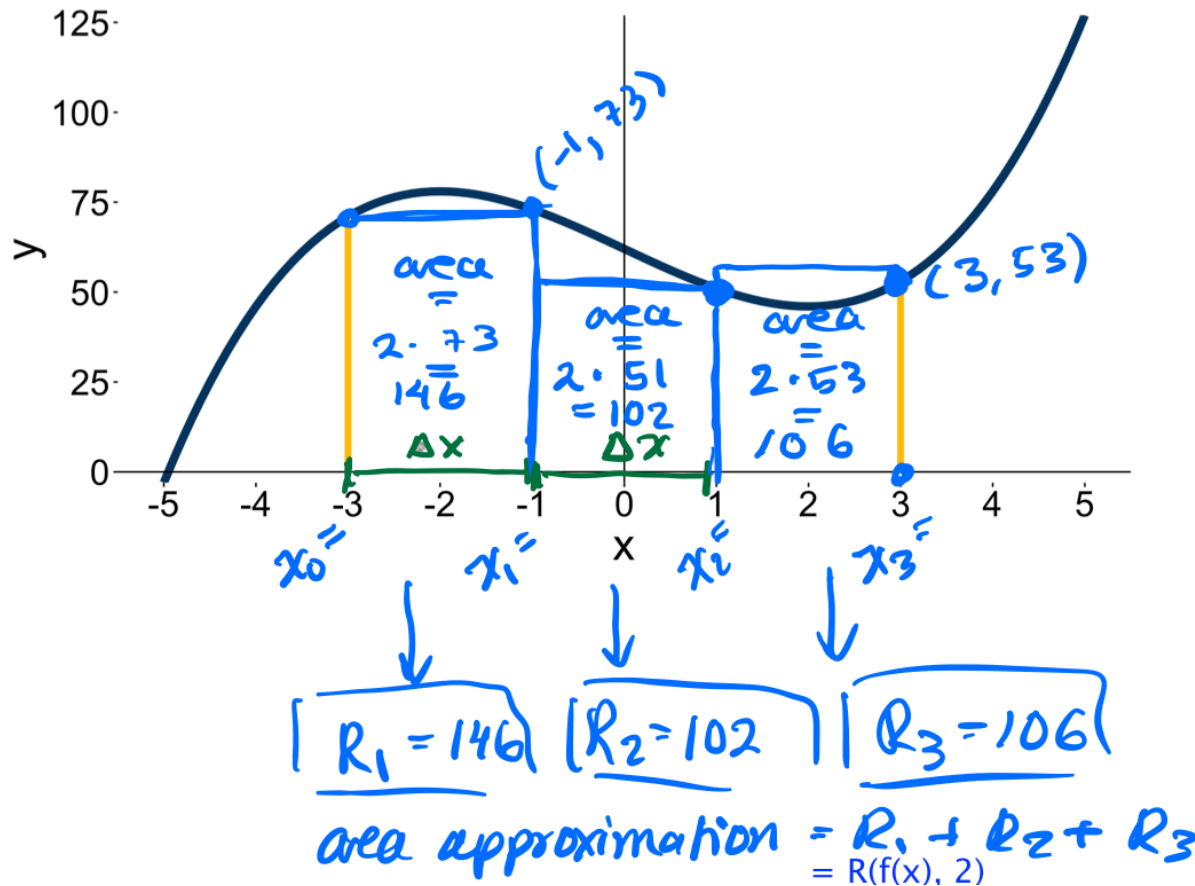
x	y
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$$x_0 = -3 \quad 71$$

$$x_1 = -1 \quad 73 = f(x_1)$$

$$x_2 = 1 \quad 51 = f(x_2)$$

$$x_3 = 3 \quad 53$$



Riemann sums

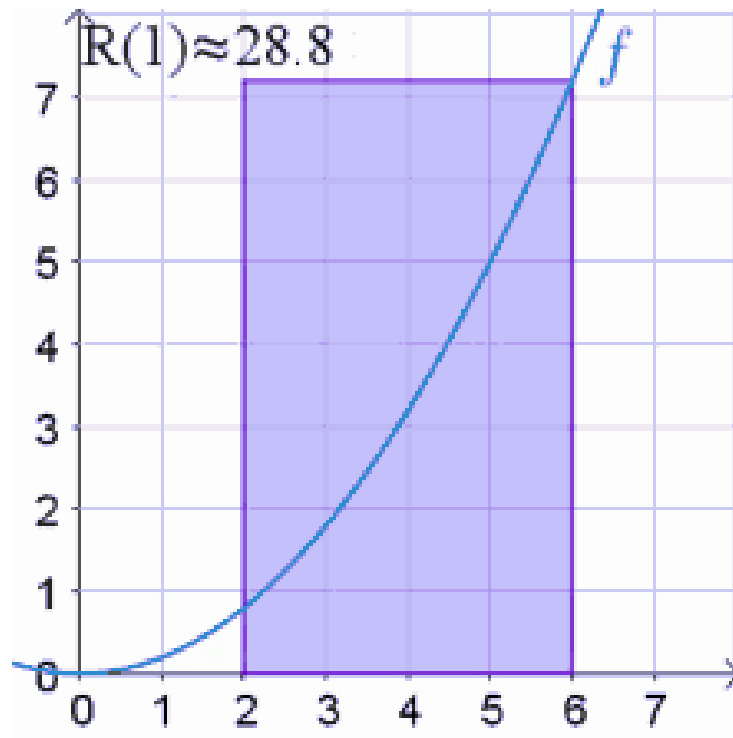
1. Divide the interval $[a, b]$ into k subintervals of width Δx .
2. The endpoints of those intervals will give you a series of x values: $x_0, x_1, \dots, x_k$.
3. Multiply $f(x_1)\Delta x$ to get the the area of the first rectangle.
4. Repeat the same steps above, now starting with x_2 as the bottom right corner of the next rectangle. Repeat until you cover the area with rectangles of width Δx .
5. Sum up the area of all k rectangles.

$$R(f(x), \Delta x) = \underbrace{f(x_1)\Delta x + f(x_2)\Delta x + \dots + f(x_k)\Delta x}_{\text{Sum of area of rectangles with width } \Delta(x)}$$

This is called a **Riemann Sum**.

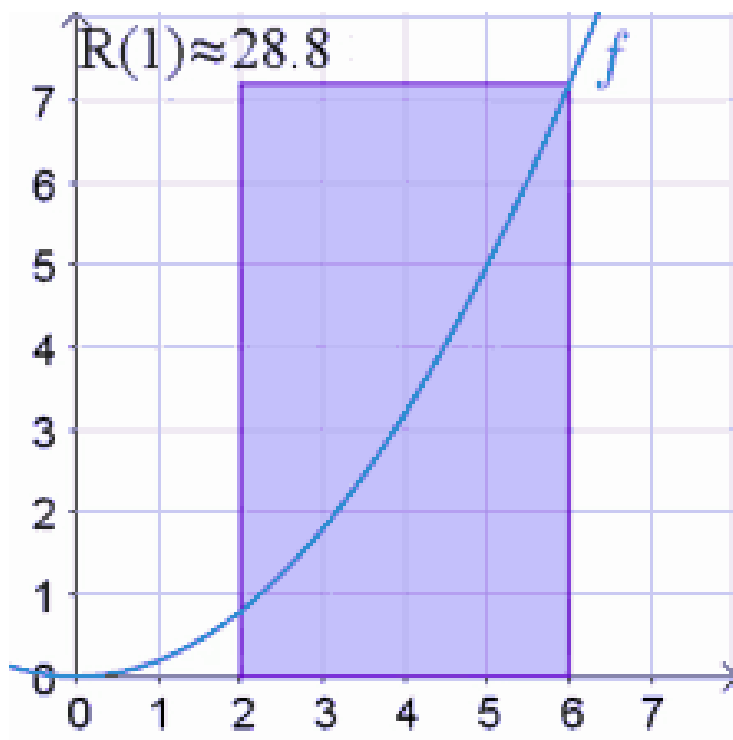
Finer and finer Riemann sums

- What if we try to make Δx (the width of the rectangles) really small? So we let $\Delta x \rightarrow 0$. This will make the number of rectangles under the curve go to infinity.
- Our estimation of the area will get better and better with more rectangles.



Finer and finer Riemann sums

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- Our estimation of the area will get better and better with more rectangles.



Take the area calculation we did before, but let $\Delta x \rightarrow 0$

$$\begin{aligned}\text{True Area} &= \lim_{\Delta x \rightarrow 0} R(f(x), \Delta x) \\ &= \int_a^b f(x) dx\end{aligned}$$

We formally call this **the integral of $f(x)$ from a to b with respect to x .**

How do we calculate integrals?

- Taking infinite sums is impossible by hand
- Luckily, integrals and derivatives are related in a very useful way:

If we integrate a derivative, we should get the same function back as the original vice-versa:

Differentiation is the inverse of integration :

Integration is the inverse of differentiation :

How do we calculate integrals?

- Taking infinite sums is impossible by hand
- Luckily, integrals and derivatives are related in a very useful way:

If we integrate a derivative, we should get the same function back as the original vice-versa:

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x) \quad \text{Differentiation is the inverse* of integration}$$

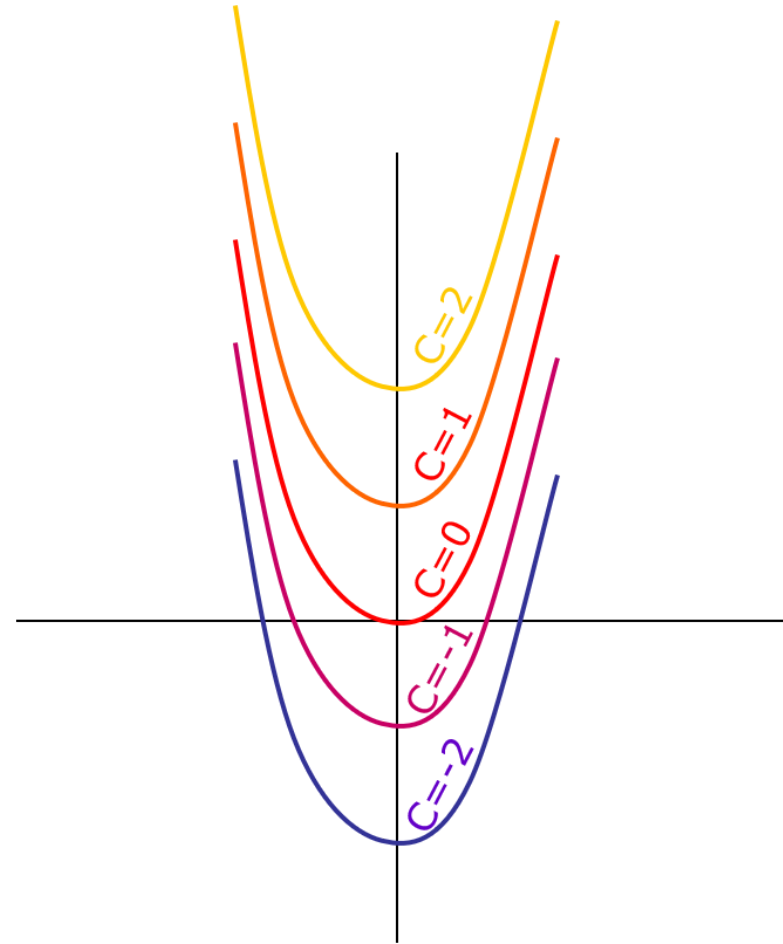
$$\int f'(x) = f(x) + C \quad \text{Integration is the inverse* of differentiation}$$

To find the integral of $\int f(t)dt$ we need to find a function $F(x)$ whose derivative is $f(x)$.

*The conditions for these to hold are more nuanced (see [Fundamental theorem of calculus](#)). But for our practical purposes we can think about it this way.

Where did that C come from?

Example



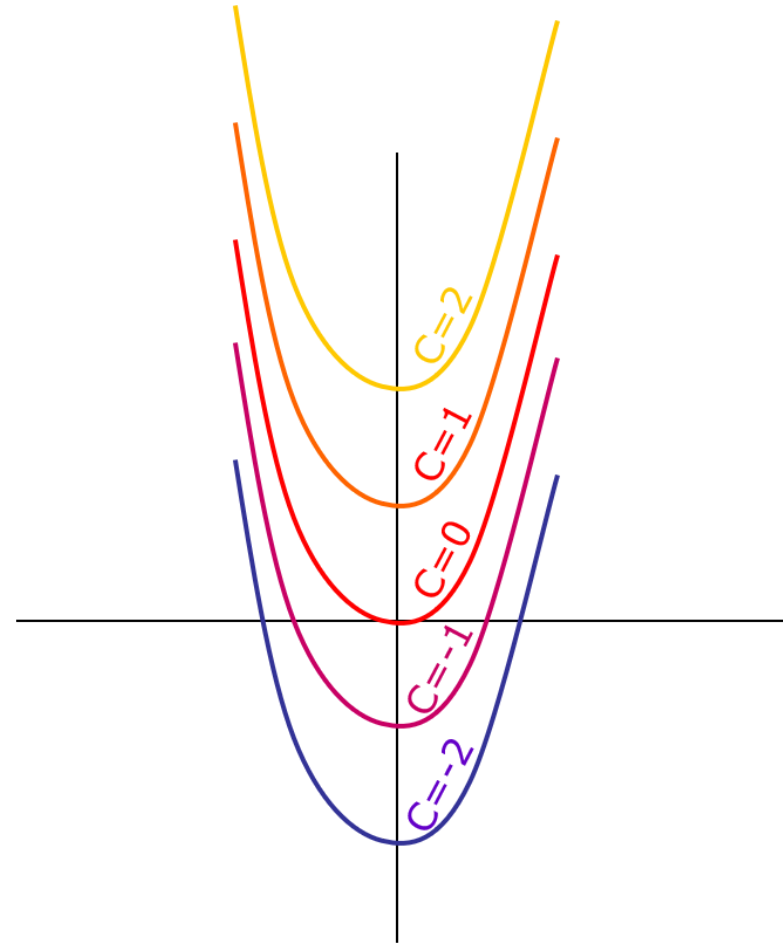
Where did that C come from?

Example

Find $\int 2x - 4$.

To find $\int 2x - 4$ we need to find a function whose derivative is $2x - 4$. What could it be?

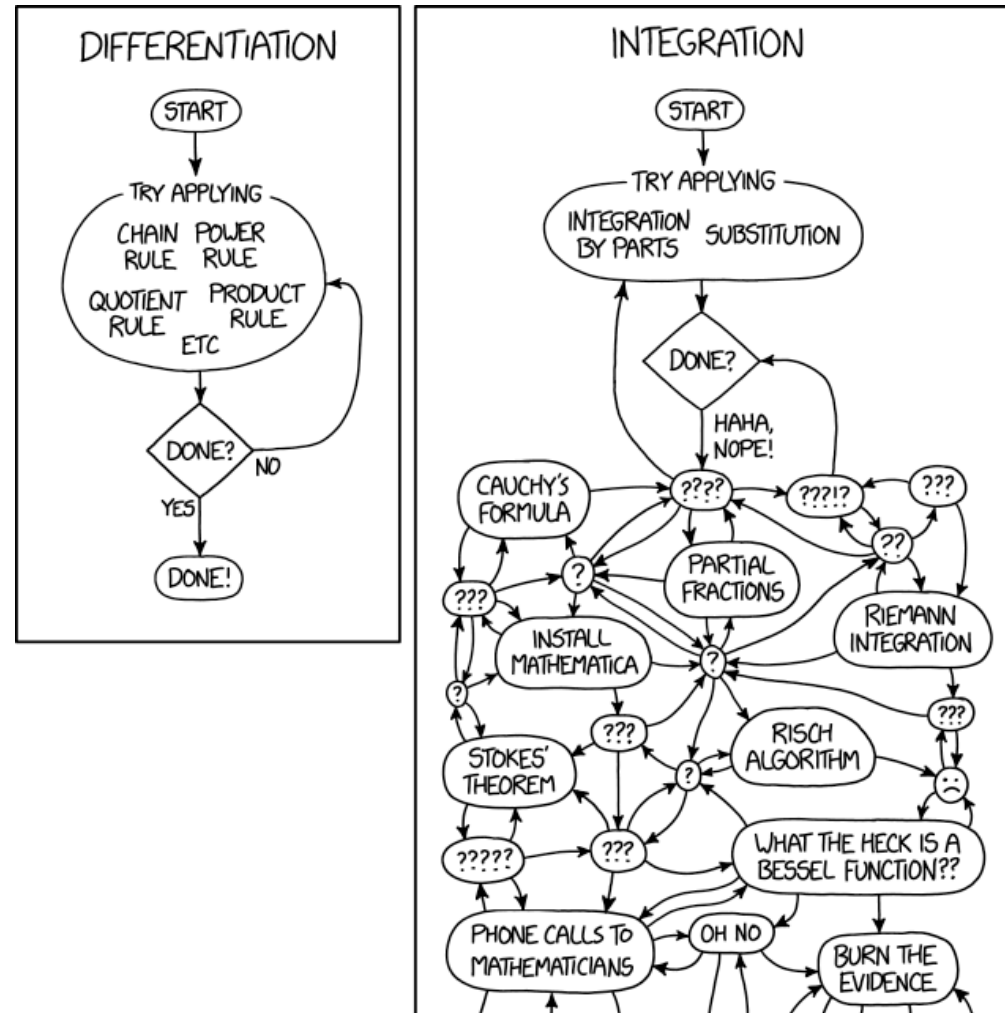
- If we integrate, we'll have $\int 2x - 4 = x^2 - 4x + C$.
- There is no way of knowing what the C should be without additional information
- C is the x -intercept and we would have to solve for it with an initial value problem.



What is integration useful for?

- Convert from rate of change to total values!
- Solving differential equations
 - i.e $\frac{dy}{dx} = 2x - 4$
- Calculating the area under a curve
- We'll see some applied exercises too

1. What am I trying to solve?
2. Does it make sense to take an integral?



Source [xkcd comics](#)

Rules of Integration

Notation

When you take an integral, the function in it can be in terms of any variable. This is just notation and it does not affect the result of the integration:

$$\int f(x)dx = \int f(t)dt = \int f(u)du.$$

Integration rules (1)

Sum and Difference Rules



Integration rules (1)

Sum and Difference Rules

The integral of a sum is equal to the sum of the integrals:

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

The integral of the difference is equal to the difference of the integrals:

$$\int [f(x) - g(x)]dx = \int f(x)dx - \int g(x)dx$$

Integration rules (2)

Constant Rule



Integration rules (2)

Constant Rule

When a function is multiplied by a constant, the constant can be taken out to multiply the integral:

$$\int cf(x)dx = c \int f(x)dx$$

Integration rules (3)

Power Rule

$\int x^n dx$ is a function whose derivative is x^n .

What could $\int x^n dx$ be?



Integration rules (3)

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Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

Examples

 Evaluate the following integrals.

a. $\int 4dx$

b. $\int 2x^2 + e^x dx$

Exercise

 Evaluate the following integral:

$$\int x^3 - 4x dx$$

Example & exercise solutions

a. $\int 4dx = 4x + C$

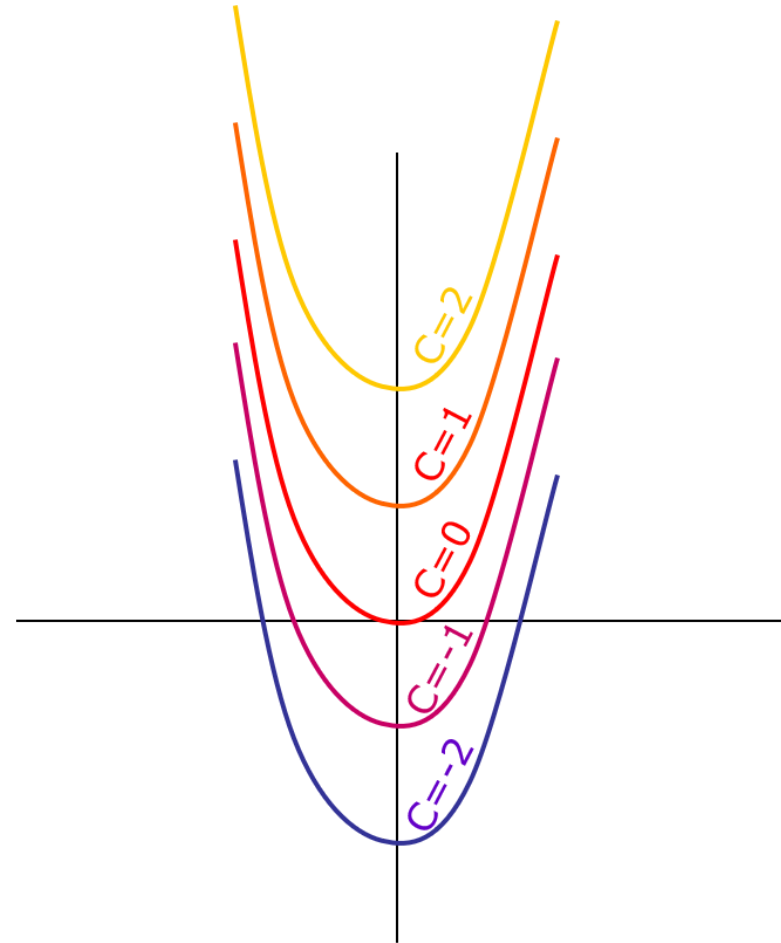
b. $\int 2x^+ e^x dx = \frac{2}{3}x^3 + e^x + C$

c. $\int x^3 - 4x dx = \frac{x^4}{4} - 2x^2 + C$

Initial value problems

Initial value problems

- To find the C term in the integral, we need extra information
- Often times that comes from being given an *initial value*
- This means being given some value of the function, e.g. $y(0) = a$ or $y(12) = b$
- We use this information to solve for what C should be



Example

 The marginal cost of producing x units of a product is:

$$\frac{dC}{dx} = 25 - 0.02x$$

Where C is the cost (in dollars), and x is the number of units produced. Given that producing 2 units of product costs \$10, what is the complete cost equation?

Example

 The marginal cost of producing x units of a product is:

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Where C is the cost (in dollars), and x is the number of units produced. Given that producing 2 units of product costs \$10, what is the complete cost equation?

We can integrate to find the original cost function C :

$$C(x) = \int \frac{dC}{dx} dx = \int 25 - 0.02x.$$

Then:

$$\int 25 - 0.02x = 25x - \frac{.02x^2}{2} + D \quad \text{Solve with Power Rule}$$

$$10 = 25(2) - \frac{0.02(2)^2}{2} + D \quad \text{Sub in } x = 2$$
$$-39.96 = D$$

$$C(x) = 25x - \frac{0.2x^2}{2} - 39.96$$

Let's take a 10 minute break



image: Flaticon.com

Exercise

A model for the rate of change in ozone concentrations over time between 1962-1984 is given by $\frac{dC}{dt} = 2t + 20$, where C is the ozone concentration (ppm) and t is the elapsed time in years since 1962. Given that in 1964 the ozone concentration was 30 ppm, what was the ozone concentration in 1982?



Exercise

A model for the rate of change in ozone concentrations over time between 1962-1984 is given by $\frac{dC}{dt} = 2t + 20$, where C is the ozone concentration (ppm) and t is the elapsed time in years since 1962. Given that in 1964 the ozone concentration was 30 ppm, what was the ozone concentration in 1982?

Given:

$$\frac{dC}{dt} = 2t + 20$$

Integrate:

$$\int dC = \int (2t + 20) dt$$

$$C(t) = t^2 + 20t + D$$

Initial condition:

$$C(2) = 30 \text{ (since 1964 is 2 years after 1962)}$$

$$30 = 2^2 + 20 \cdot 2 + D$$

$$D = 30 - 44 = -14$$

Thus:

$$C(t) = t^2 + 20t - 14$$

Evaluate at $t = 20$ (1982):

$$\begin{aligned} C(20) &= 20^2 + 20 \cdot 20 - 14 \\ &= 400 + 400 - 14 \\ &= 786 \text{ ppm} \end{aligned}$$

Definite integrals

Indefinite vs. definite integrals

- We have been working with **indefinite integrals**.

$$\int f(x)dx$$

This means there are no integration bounds and we need initial values to find integration constants.



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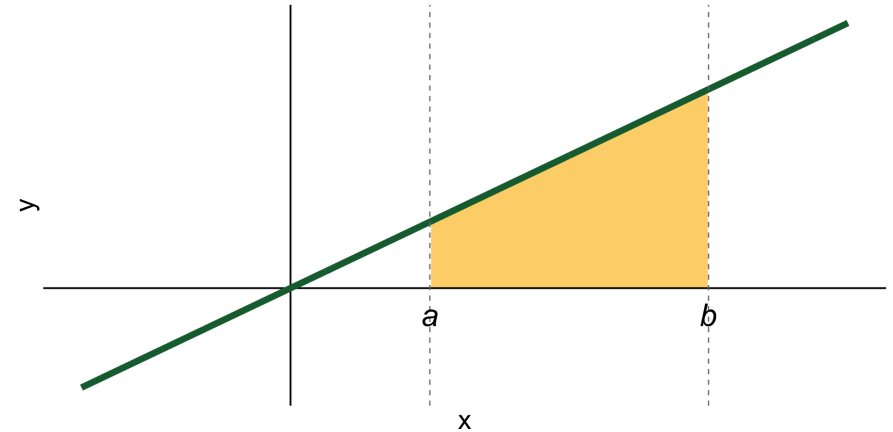
- Definite integrals** have start and end values a and b :

$$\int_a^b f(x)dx.$$

To find the area under the curve along an interval $[a, b]$, evaluate the antiderivative at the endpoints and subtract them.

 **Example**

$$\begin{aligned}\int_a^b xdx &= \frac{1}{2}x^2 \Big|_a^b \\ &= \frac{1}{2}(b)^2 - \frac{1}{2}(a)^2\end{aligned}$$



Example

 Evaluate the following integral.

$$\int_{-1}^2 3x^2 dx$$

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We have that:

$$\begin{aligned}\int_{-1}^2 3x^2 dx &= x^3 \Big|_{-1}^2 \\ &= (2)^3 - (-1)^3 \\ &= 9\end{aligned}$$

Exercise

Integrate

$$\int_2^4 \frac{1}{2}x.$$

Make a diagram to represent the area this integral represents.



Solution

We get that

$$\begin{aligned}\int_2^4 \frac{1}{2}x \, dx &= \frac{1}{2} \int_2^4 x \, dx \\&= \frac{1}{2} \left[\frac{x^2}{2} \right]_2^4 \\&= \frac{1}{4} (4^2 - 2^2) \\&= \frac{1}{4} (16 - 4) \\&= 3\end{aligned}$$

Differential equations

What is a differential equation?

A **differential equation** is any equation which contains derivatives.

The goal of the differential equation is to find a function that satisfies the equation.

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A **differential equation** is any equation which contains derivatives.

The goal of the differential equation is to find a function that satisfies the equation.

Example

Solve for $y(x)$ in $y'(x) = y(x) + x$.

Notation is often simplified as

Solve for y in $y' = y + x$.

- Is a differential equation because we are trying to find the function y with information about its derivative y' .
- Translated as: find a function $y(x)$ whose derivative equals the function plus x .

What is a solution to a differential equation?

In a “normal” equation the solutions are a set of values of x .

Solve for x in $x^2 - 4 = 0$.

How would you check that $x = 2$ is a solution for the equation?

In a differential equation, the solutions are whole functions.

Solve for y in $y' = y + x$.

How would you check that $y(x) = -x - 1$ is a solution for this differential equation?

Differential equations: vocab

*Sometimes functions have more than one variable and we can take **partial derivatives** with respect to each variable.*

- **Ordinary differential equation (ODE):** Does not contain partial derivatives

$$\frac{df}{dt} = 3.2 - f(t)$$

- **Partial differential equations (PDE):** Differential equations with partial derivatives

Order of an ODE

Order: The order of a differential equation is the highest order for any differential expression in the equation

Example:

$\frac{df}{dt} = 3.2 - f(t)$ is a **first order ordinary differential equation**

Differential equations help us understand changing environments

- For some natural phenomena it can be easier to describe them in terms of derivatives than give an explicit function.
- By specifying how something changes, we are describing a process or behavior over time.
- Examples:



Describing population dynamics



Groundwater transport of absorbed contaminants

Example: Lotka-Volterra (predator-prey) equations

Prey: $\frac{dN}{dt} = rN - \alpha NP$

Predator: $\frac{dP}{dt} = abNP - mP$

Where:

- N is number of prey (e.g. rabbits)
- P is number of predators (e.g. wolves)
- r, α, a, b, m are positive parameters

What do the different pieces of the equation *mean*?

Interpretation of prey equation:

$$\frac{dN}{dt} = rN - aNP$$

The pieces:

- $\frac{dN}{dt}$: Rate of growth / decline in prey abundance
- rN : Population growth (r is the intrinsic rate of increase)
- $-aNP$: Population loss due to predation (where a is the attack rate, or capture efficiency, of the predator)

Interpretation of predator equation:

$$\frac{dP}{dt} = baNP - mP$$

The pieces:

- $\frac{dP}{dt}$: Rate of growth / decline in predator population
- $baNP$: Predator population growth
- $-mP$: Predator population loss (where m is the predator mortality)

Or, in pictures:

$$\begin{aligned}\frac{d \text{🐰}}{dt} &= r \text{🐰} - \alpha \text{🐰} \text{🐺} \\ \frac{d \text{🐺}}{dt} &= \beta \text{🐰} \text{🐺} - q \text{🐺} \text{🐺}\end{aligned}$$

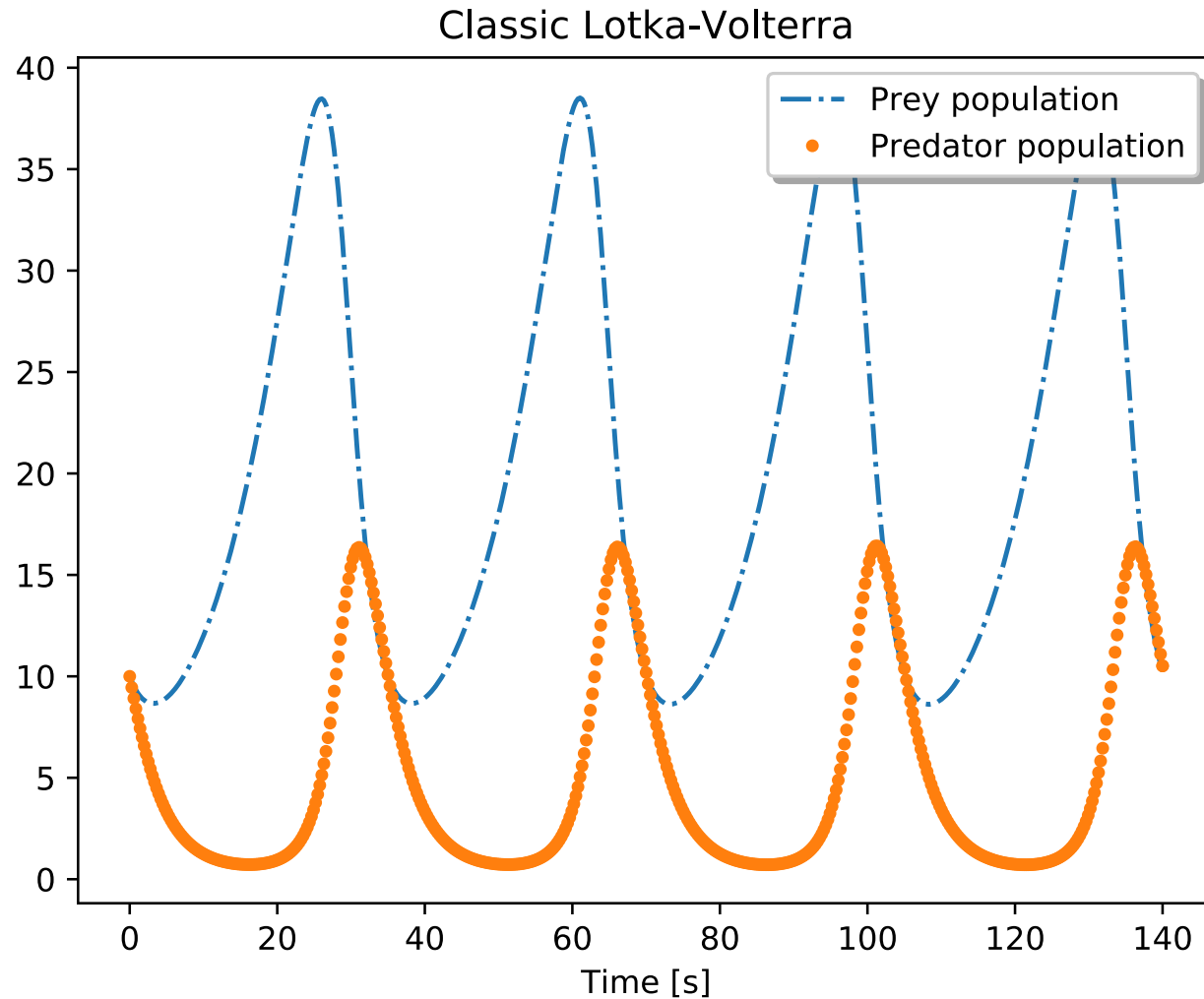
Exponential growth

Gets eaten by wolves

Increases with more food

Decreases with competition

Finding $N(t)$ and $P(t)$?



Some differential equations can be solved analytically:



Some differential equations can be solved analytically:

$$\frac{dy}{dx} = y$$

Separate variables and integrate both sides:

$$\int \frac{1}{y} dy = \int 1 dx$$

Yielding:

$$\ln(y) = x \text{ or } y = e^x$$

Solving differential equations numerically

Find *approximate* solutions to differential equations when finding an analytical solution would be really challenging (...which is pretty often).

Instead, computers can numerically approximate solutions by predicting nearby values based on the *slope*.

There are many methods for solving differential equations numerically. You'll do it programmatically later on.

Wrapping up...

These materials are an iterative effort



Allison Horst



Ruth Oliver



Nathan Grimes



Carmen Galaz García



Sam Shanny-Csik



Alessandra Vidal Meza



Cella Schnabel

We covered a lot!



@allison-horst

Pleaes fill out the exit survey!



<https://bit.ly/4yfEknE>



Have a great Fall Quarter!

